BHARATH KISHORE REDDY TIMMAPURAM

Data Engineer

Houston, TX, USA | +1-832-602-4262 | bharath.tbr191@gmail.com | LinkedIn

---

PROFESSIONAL SUMMARY

Data Engineer with **4+ years** of experience delivering scalable **data pipelines, analytics platforms** and **business intelligence solutions** across finance, retail, and e-commerce domains. Strong expertise in **Python, SQL, Apache Spark, Airflow** and cloud data warehouses **(AWS Redshift, Snowflake, ADF)** for building reliable **ETL** systems and analytics-ready datasets. Proven ability to bridge data engineering and analytics, enabling stakeholders with high-quality dashboards, KPIs, and insights using **Power BI** and **Tableau**. Adept at data governance, automation, and cross-functional collaboration to drive data-driven decision-making.

TECHNICAL SKILLS

Programming & Querying:	Python (Pandas, NumPy, PySpark, Keras, Tensorflow), SQL (MySQL, PostgreSQL), Shell Scripting, R, java, Angular.
Data Engineering & Big Data:	Apache Spark, Airflow, AWS Glue, ETL/ELT Pipelines, Data Modeling
Cloud & Platforms:	AWS (S3, Redshift, Lambda, Athena), Azure Data Lake, Snowflake
Analytics & BI:	Power BI (DAX, Power Query), Tableau, Excel (Pivot Tables, Macros)
DevOps & Automation:	Git, Jenkins, CI/CD, Monitoring & Alerting, Jira.
Data Governance & Quality:	Data Validation, Anomaly Detection, Data Cleansing, Compliance

---

PROFESSIONAL EXPERIENCE

Apex Fintech Solutions – Texas, USA

Data Engineer **January 2024 – Present**

Project: Retail & E-Commerce Transaction Analytics

- Architected and automated **end-to-end cloud-native ETL pipelines** using **Python,R, Apache Spark, Airflow, and SQL**, ingesting high-volume transactional data from **APIs, AWS S3, Snowflake, and MySQL**, reducing manual ETL effort by **40%**.
- Designed and optimized **analytics-ready data warehouses** in **AWS Redshift and Snowflake**, applying **partitioning, clustering, indexing, and query tuning**, improving query performance and dashboard response times by **50%**.
- Implemented **robust data quality and governance frameworks** using Python-based validation, schema enforcement, and Airflow DAG checks, achieving **99% data consistency** across millions of daily transaction records.
- Built **dimensional and normalized data models (fact & dimension tables)** to support executive dashboards, KPI tracking, and **ML-ready analytical layers** for revenue forecasting and anomaly detection.
- Enabled **near real-time analytics** by implementing incremental loading strategies and optimized transformation logic, reducing end-to-end data latency from **4 hours to under 1 hour**.
- Established **CI/CD pipelines** using **Jenkins and Git** for automated deployment, version control, and monitoring of Airflow workflows, reducing failure recovery time by **60%**.
- Integrated **pipeline monitoring and alerting** via Airflow and AWS CloudWatch, enabling proactive issue detection and improving system reliability and uptime.
- Delivered **analytics-ready datasets** powering **Power BI and Tableau dashboards** for transaction trends, customer behavior, revenue performance, and operational KPIs.
- Collaborated cross-functionally with **product, analytics, and business stakeholders** to translate reporting requirements into scalable data models, metrics, and trusted single-source-of-truth datasets.
- Contributed to **technical documentation, best practices, and reusable pipeline templates**, improving onboarding efficiency and standardizing data engineering workflows.

Deloitte – India

Associate – Data & Analytics **May 2020 – July 2022**
Project: Banking Transaction Analytics & Performance Optimization

- Designed and developed **scalable ETL pipelines** using **Azure Data Factory (ADF), SSIS, SQL Server, and Python** to consolidate high-volume banking transactional data from **core banking, CRM, and payment systems**, processing **10M+ records daily**.
- Implemented **incremental and parameterized data loads**, reducing ETL execution time by **45%** and enabling **near real-time availability** of transactional and SLA metrics.
- Built **high-performance data models** using **star schema and normalized designs**, applying **partitioning, indexing, and surrogate keys** to improve query performance and analytics scalability.
- Optimized **complex SQL queries, stored procedures, and workflows**, reducing reporting and dashboard refresh times by **40%**.
- Developed **data quality, validation, and reconciliation frameworks** using **Python and SQL** to enforce schema consistency, detect duplicates, and ensure **98% data accuracy** for regulatory and audit reporting.
- Delivered **interactive Power BI dashboards** using **DAX and Power Query** to track transaction volume, failure rates, processing latency, SLA adherence, and operational exceptions.
- Automated **report refreshes, anomaly detection, and exception alerts**, reducing manual reporting effort by **65%**, improving SLA compliance by **25%**, and enabling **real-time and near real-time analytics** through optimized refresh strategies and **DirectQuery-based reporting**.
- Collaborated with **finance, operations, IT, and compliance teams** to translate business and regulatory requirements into **governed, auditable analytics solutions**, while supporting **ad-hoc analysis, root-cause investigations, and forecasting**.
- Implemented **monitoring and alerting mechanisms** using **Azure Monitor and automated scripts**, and created **technical documentation and reusable templates**, improving system reliability, onboarding efficiency, and analytics standardization.

EDUCATION

Master's in Computing and Data Science
Sam Houston State University, USA
Aug 2022 – Dec 2023